

Variables in Vb.net

Instructor Name: Assist.Lec. Aseel F. Alshaibanee

Instructor Name: Assist.Lec. Marwa Ali

Département: Computer Science

Level: Two

What is Variable?

- A **variable** is nothing but a **name** given to a **storage area** that our programs can manipulate. Each variable in VB.Net has a **specific type**, which **determines the size and layout of the variable's memory**.
- You often have to **store values** when you perform **calculations** with Visual Basic. For example, you might want to **calculate several values, compare them, and perform different operations on them**.

Common data types

- Integer → Integers
- Double → Decimals
- String → Texts
- Boolean → True/False
- Char → Single character

Variable Declaration in VB.Net

- **Dim** variableName **As** DataType
- Example:

Dim StudentID **As** Integer

Dim StudentName **As** String

Dim Salary **As** Double

Dim count1, count2 **As** Integer

Variable Initialization in VB.Net

- Variables are initialized (assigned a value) with an **equal** sign followed by a **constant expression**. The general form of initialization is.

`variable_name = value`

Example:

```
Dim pi As Double
```

```
pi = 3.14159
```

Variable Initialization in VB.Net

- You can **initialize a variable at the time of declaration** as follows:

```
Dim StudentID As Integer = 100
```

```
Dim StudentName As String = "Bill Smith"
```


Example1

```
Private Sub Button1_Click(sender...
```

```
    Dim a As Integer = 12
```

```
    Dim b As Integer = 6
```

```
    Dim sum As Integer
```

```
    sum = a + b
```

```
    TextBox1.Text = sum
```

```
End Sub
```

Example2

- Declares a **variable** of type **String** to store a **student's name**.
- Declares **a variable** of type **Integer** to store the **student's age**.
- Prints a sentence in the following format:

My name is Ali and I am 20 years old.

Example2

- Solve:

```
Private Sub Button2_Click(sender As ...
```

```
    Dim studentName As String = "Ali"
```

```
    Dim studentAge As Integer = 20
```

```
    ' Print the sentence
```

```
    MessageBox.Show("My name is " & studentName & " and I am " & studentAge & " years old.")
```

```
End Sub
```

CONSTANTS

- In VB.NET, a **constant** is a **fixed value** that does not change during the execution of a program. **Constants are declared using the Const statement and must be assigned a value at the time of declaration.**
- Constants can be of any of the basic data types like an **integer constant, a floating constant, a character constant, or a string.**

CONSTANTS

- The Constant declaration are as following:

Const ConstName As Datatype

Example:

Const pi As Double

Pi=3.14

Or

Const pi As Double =3.14

Thank You